



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- Events A and B are such that $P(A \cup B) = 0.75$, $P(A \cap B) = 0.25$ and $P(\bar{A}) = \frac{2}{3}$. Find $P(B)$.
- From a pack of 52 cards, two are drawn at random. Find the probability that one is a king and the other a queen.
- Explain the concept of random variable. What is the distribution function and what are its properties?
- What are the four properties that must be present in order to use binomial distribution?
- Mark has 4 \$. He bet 4 times for 1\$. What is the probability that he breaks even.
- Differentiate between independent and mutually exclusive events. Are independent events mutually exclusive?

Answer the following questions.

Q.2. The probability that a person chosen has a certain disease is 0.03. Medical diagnostic tests are available to determine whether the person actually has the disease. If the disease is actually present, the probability that the medical diagnostic test will give a positive result is 0.90. If the disease is not actually present, the probability of a positive test result is 0.02. Suppose that the medical diagnostic test has given a positive result. What is the probability that the disease is actually present? What is the probability of positive test result?

(10)

Q.3. Compute the mean, variance, coefficient of skewness and kurtosis of Poisson distribution.

(10)

Q.4.a) Show that the hypergeometric distribution $h(x; N, n, k)$ can be approximated by the binomial distribution for large N and k .

- An insurance salesman sells policies to 5 men, all of identical age and in good health. According to the actuarial tables, the probability that a man of this particular age will be alive 30 years hence is $\frac{2}{3}$. Find the probability that in 30 years (i) at least 3 men, (ii) at most one man will be alive.

(5+5)